

Ffowcs Williams and Hawkings (FW-H) equation

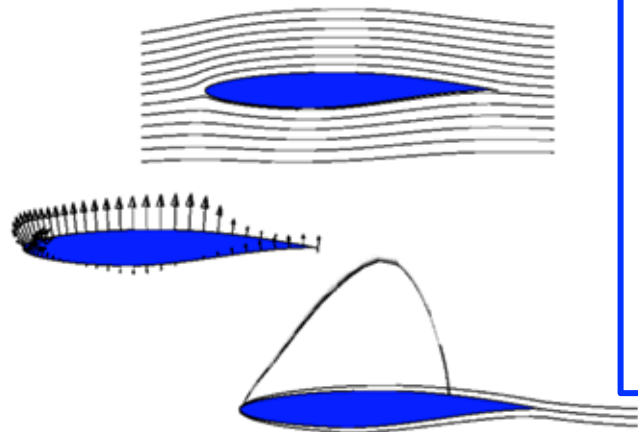
- ❑ So far, we have derived the general solution to the wave equation for a medium with stationary scattering surfaces
- ❑ **What happens when the surface is moving?**



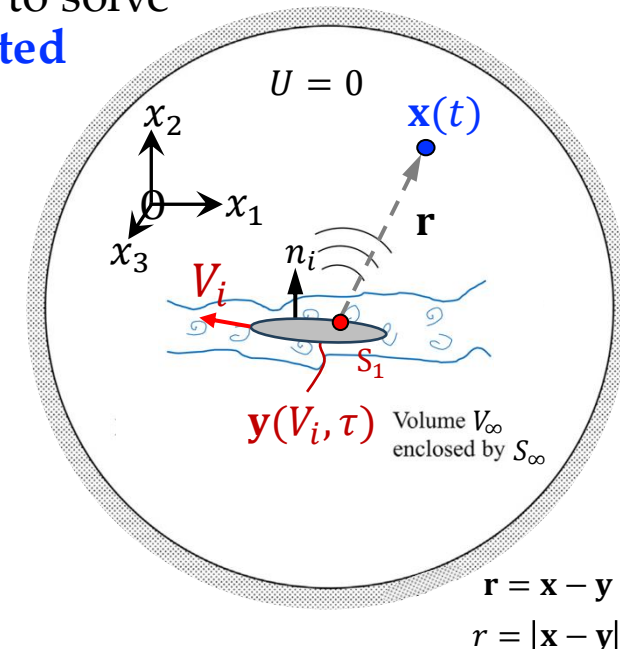
Aircraft Aeroacoustics

Ffowcs Williams and Hawkings (FW-H) equation

- ❑ In many aeronautical applications, the surfaces/bodies are moving (e.g., propellers and rotors, aircraft translation motion), so we need to modify our analysis to take full account of surface motion
- ❑ Powerful techniques to address this problem have been pioneered by [Ffowcs Williams and Hawkings](#)¹ (1969), as well as [Farassat](#)² (1981)
- ❑ Derivation is based on the concept of **generalized derivatives**, which is used to solve Lighthill's equation for a medium that includes **moving surfaces and convected turbulent flow**



$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 H_s = & \frac{\partial}{\partial t} \int_{S_0} \left[\frac{(\rho v_j - \rho' V_j) n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS \\ & - \frac{\partial}{\partial x_i} \int_{S_0} \left[\frac{(p_{ij} + \rho v_i (v_j - V_j)) n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS \\ & + \frac{\partial^2}{\partial x_i \partial x_j} \int_{V_0} \left[\frac{T_{ij}}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dV(\mathbf{z}) \end{aligned}$$



SOUND GENERATION BY TURBULENCE AND SURFACES IN ARBITRARY MOTION

By J. L. FFWCS WILLIAMS AND D. L. HAWKINGS
Department of Mathematics, Imperial College, London
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Linear Acoustic Formulas for Calculation of Rotating Blade Noise

F. Farassat

NASA Langley Research Center, Hampton, Va.

A suitable approach to the problem of sound generation by rotating blades is presented. The problem is solved by means of the generalized derivatives of the surface motion. The resulting formulas are presented in a form suitable for numerical calculation. The results are compared with those of the Ffowcs Williams and Hawkings (1969) and Farassat (1981) formulas. It is shown that the generalized derivatives approach is more general and more accurate than the other two approaches. The results are presented in a form suitable for numerical calculation. The results are compared with those of the Ffowcs Williams and Hawkings (1969) and Farassat (1981) formulas. It is shown that the generalized derivatives approach is more general and more accurate than the other two approaches.

¹ Ffowcs Williams, J. E., and Hawkings, D. L. (1969) "Sound generation by turbulence and surfaces in arbitrary motion", *Phil Trans A*, 264(1151), pp. 321–342.

² Farassat, F. (1981) "Linear acoustic formulas for calculation of rotating blade noise", *AIAA Journal*, 19(9), pp. 1122–1130.

Generalized derivative - review

- ❑ A **generalized derivative** extends the concept of an ordinary derivative to **discontinuous functions**
- ❑ Consider the Heaviside step function $H_s(x)$:

$$H_s(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x < 0 \end{cases}$$

- ❑ With its derivative being: $\frac{\partial H_s(x)}{\partial x} = \begin{cases} 0, & \text{if } x > 0 \text{ and } x < 0 \\ \infty, & \text{if } x = 0 \end{cases}$

- ❑ The integral of the derivative is given by:

$$\int_{x_1}^x \frac{\partial H_s(x')}{\partial x'} dx' = \begin{cases} 1, & \text{if } x > 0 \text{ and } x_1 < 0 \\ 0, & \text{if } x < 0 \text{ and } x_1 < 0, \text{ or } x > 0 \text{ and } x_1 > 0 \end{cases}$$

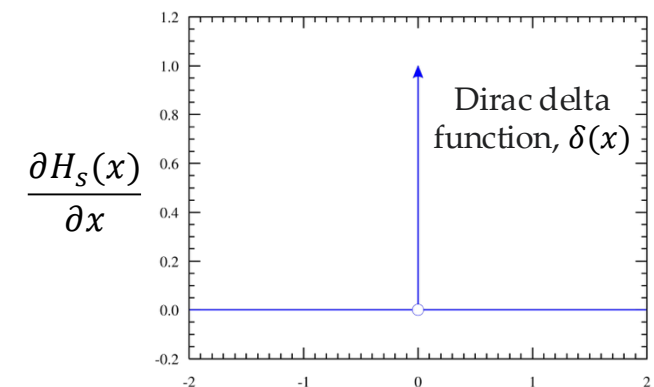
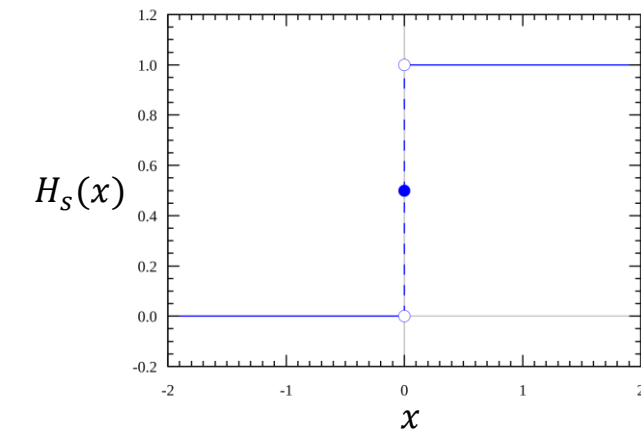
- ❑ **Looks familiar?**

➤ Derivative of $H_s(x)$ has properties of *Dirac delta function*!

$$\delta(x - x_0) = \begin{cases} \infty, & \text{if } x = x_0 \\ 0, & \text{if } x \neq x_0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x - x_0) dx = \begin{cases} 1, & \text{if } x = x_0 \\ 0, & \text{if } x \neq x_0 \end{cases}$$

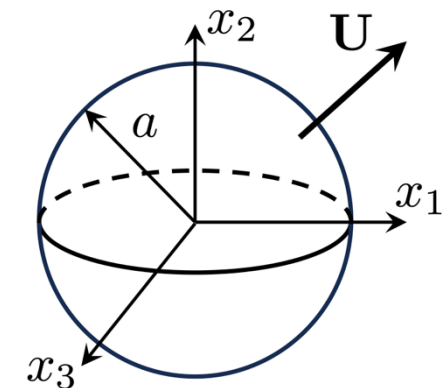
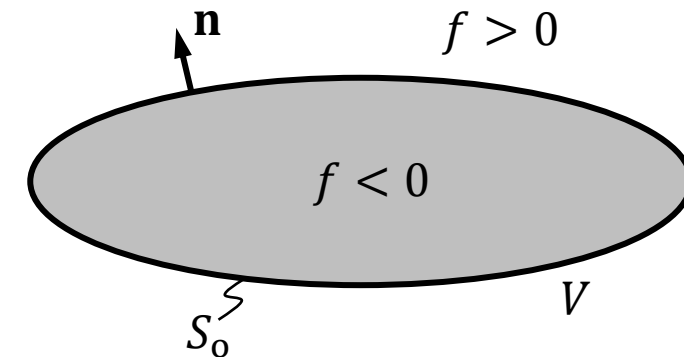
$$\delta(x - x_0) = \frac{\partial H_s(x - x_0)}{\partial x}$$



Generalized derivative - review

- ❑ We can extend this concept to surfaces that bound a region
- ❑ Consider a function $f(\mathbf{x})$ that is greater than zero outside a volume V enclosed by a surface S_0 and less than zero outside – therefore, surface defined by $f = 0$
 - Spherical surface of radius a : $f = x_1^2 + x_2^2 + x_3^2 - a^2$
 - Cylinder of radius a : $f = x_1^2 + x_2^2 - a^2$
- ❑ Unit normal vector pointing out of surface: $\mathbf{n} = \frac{\nabla f}{|\nabla f|}$ @ $f = 0$
- ❑ **Surface description is general** and can be extended to **moving surfaces** by letting f also be a function of time – $f(\mathbf{x}, t)$
 - For example, a **sphere moving with velocity** $\mathbf{U}$ is defined by:

$$f(\mathbf{x}, t) = |\mathbf{x} - \mathbf{U}t|^2 - a^2$$

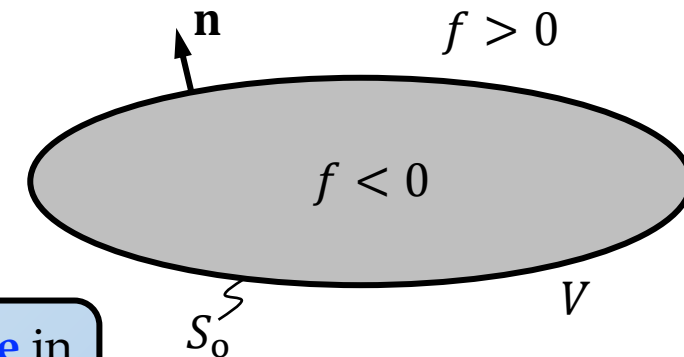


Generalized derivative – towards moving surface

- ❑ To analyze the flow field in the region exterior to a surface S_0 , which may be in **arbitrary accelerated motion**, we need information about the *flow variables* in the region exterior to S_0
 - Note that we know *nothing* about the flow variables inside S_0
- ❑ Thus, for example, **flow velocity can only be defined outside S_0**
- ❑ Introducing a *new flow velocity variable* $\mathbf{v}H_s(f)$:

$$\mathbf{v}H_s(f) = \begin{cases} \mathbf{v}, & \text{if } f > 0 \\ 0, & \text{if } f < 0 \end{cases}$$

We now have a **flow variable that is defined everywhere** in the presence of an **arbitrary moving surface**



- ❑ **Divergence** of this new function: $\nabla \cdot (\mathbf{v}H_s(f)) = H_s(f)\nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla H_s(f)$

$$\mathbf{n} = \frac{\nabla f}{|\nabla f|}$$

$$\delta(x) = \frac{\partial H_s(x)}{\partial x}$$



$$\nabla \cdot (\mathbf{v}H_s(f)) = H_s(f)\nabla \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{n}\delta(f)|\nabla f|$$

$$\nabla H_s(f) = \frac{\partial H_s}{\partial f} \nabla f = \delta(f)|\nabla f| \mathbf{n}$$

Generalized derivative – towards moving surface

- Integrate over an infinite region V_∞ bounded at infinity by the surface S_∞ , and use *divergence theorem*:

$$\int_{V_\infty} \nabla \cdot (\mathbf{v} H_s(f)) dV = - \int_{S_\infty} (\mathbf{v} H_s(f)) \cdot \mathbf{n} dS \xRightarrow{H_s(f) = 1 \text{ on } S_\infty} \int_{V_\infty} (H_s(f) \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|) dV = - \int_{S_\infty} \mathbf{v} \cdot \mathbf{n} dS \quad (\text{I})$$

$$\nabla \cdot (\mathbf{v} H_s(f)) = H_s(f) \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|$$

- For the **first term** in the volume integral, we recognized $H_s = 0$ inside the body, and so we can restrict the integration to the **exterior volume region** V_0 outside of S_0 ($f > 0$) and apply the divergence theorem:

$$\int_{V_\infty} (H_s(f) \nabla \cdot \mathbf{v}) dV = \int_{V_0} (\nabla \cdot \mathbf{v}) dV = - \int_{S_0 + S_\infty} \mathbf{v} \cdot \mathbf{n} dS \quad (\text{II})$$

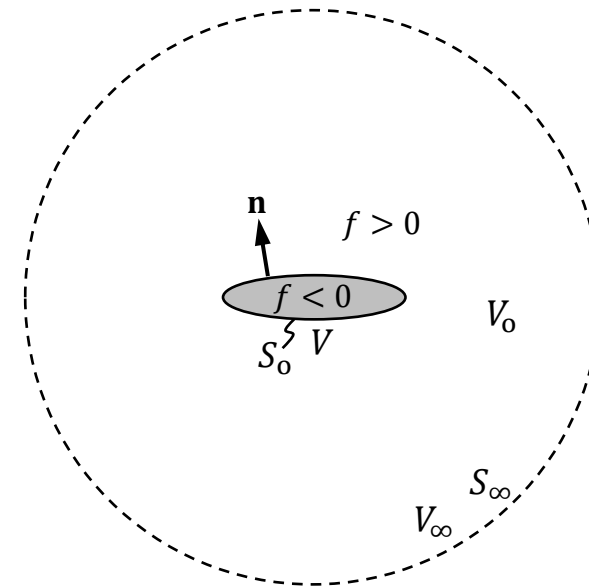
Combining (I) and (II)

$$\int_{V_\infty} (\mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|) dV = \int_{S_0} \mathbf{v} \cdot \mathbf{n} dS$$

$$\nabla H_s(f) = \delta(f) |\nabla f| \mathbf{n}$$

Key result – Dirac delta distribution $\delta(f) |\nabla f|$ turns a volume integral over all space, where the surface motion is encoded implicitly by $f(\mathbf{x}, t) = 0$, into an **equivalent** surface integral over the moving body surface $S_0(t)$

$$H_s(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x < 0 \end{cases}$$



Generalized derivative – towards moving surface

- Now, consider the time derivative of the *new flow velocity variable* $\mathbf{v}H_s(f)$:

$$\boxed{\frac{\partial(\mathbf{v}H_s(f))}{\partial t} = H_s(f) \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \delta(f) \frac{\partial f}{\partial t}}$$

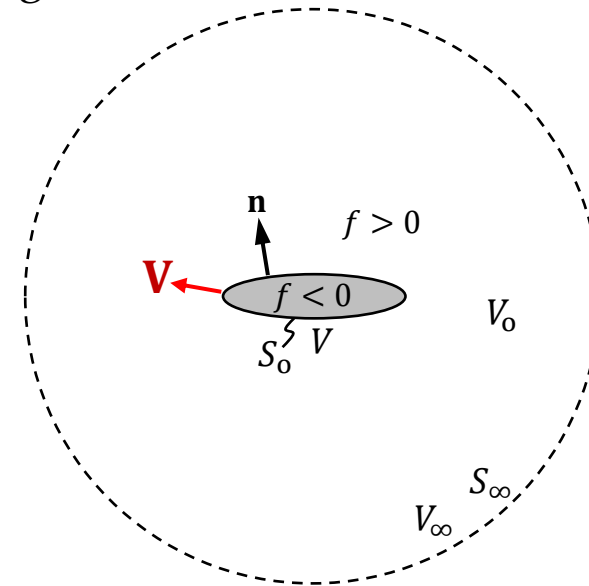
$$\delta(t) = \frac{\partial H_s(t)}{\partial t}$$

- For a surface defined by $f(\mathbf{x}, t)$ moving with velocity $\mathbf{V}$, it follows that $f(\mathbf{x} - \int \mathbf{V} dt)$ on the surface, and so the chain rule on the surface ($Df/Dt = \partial f/\partial t + \mathbf{V} \cdot \nabla f = 0$, since $f = 0$) gives:

$$\left[\frac{\partial f(\mathbf{x} - \int \mathbf{V} dt)}{\partial t} \right]_{f=0} = [-\mathbf{V} \cdot \nabla f]_{f=0}$$

$$\Rightarrow \boxed{\frac{\partial(\mathbf{v}H_s(f))}{\partial t} = H_s(f) \frac{\partial \mathbf{v}}{\partial t} - \mathbf{v} \delta(f) (\mathbf{V} \cdot \mathbf{n} |\nabla f|)} \quad \mathbf{n} = \frac{\nabla f}{|\nabla f|}$$

- So the second term on the right is specified in terms of the surface velocity $\mathbf{V}$



Generalized derivative – towards moving surface

- Similarly, for a scalar variable, such as the density perturbation ρ' , we find:

$$\frac{\partial(\rho' H_s(f))}{\partial t} = H_s(f) \frac{\partial \rho'}{\partial t} - \rho' \delta(f) (\mathbf{V} \cdot \mathbf{n} |\nabla f|)$$

$$\delta(x) = \frac{\partial H_s(x)}{\partial x}$$

$$\frac{DH_s(f)}{Dt} = \frac{\partial H_s}{\partial f} \frac{Df}{Dt} = \delta(f) \frac{Df}{Dt}$$

- Consider the material derivative of $\mathbf{v} H_s(f)$ given by:

$$\frac{D(\mathbf{v} H_s)}{Dt} = \frac{D\mathbf{v}}{Dt} H_s + \mathbf{v} \frac{Df}{Dt} \delta(f)$$

Should be 0 for a solid surface

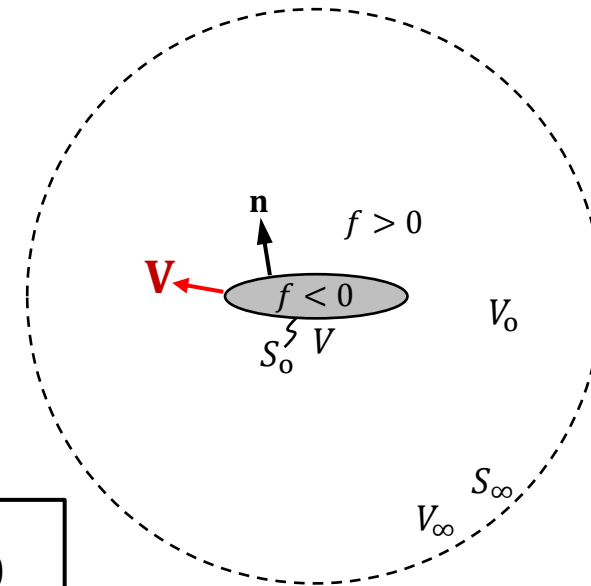
➤ **BUT!** $\frac{Df}{Dt} = \frac{\partial f}{\partial t} + (\mathbf{v} \cdot \nabla) f \implies \frac{Df}{Dt} = (\mathbf{v} - \mathbf{V}) \cdot \mathbf{n} |\nabla f|$

$\frac{\partial f}{\partial t} = -\mathbf{V} \cdot \mathbf{n} |\nabla f|$

- For an **impermeable (solid) surface** – $\mathbf{v} \cdot \mathbf{n} = \mathbf{V} \cdot \mathbf{n}$

➤ Fluid velocity normal to the surface is equal to the surface velocity

$$\frac{Df}{Dt} = 0$$



The FW-H equation

- We now show how the concept of **generalized derivatives** may be used to obtain the solution to Lighthill's equation in the presence of moving surfaces
- We start by evaluating the **continuity and momentum equations** in terms of the new variables $\mathbf{v}H_s$, pH_s , and ρH_s , which are defined everywhere in an unbounded infinite volume V_∞
- To obtain the continuity equation in terms of the new variables, we expand:

$$\frac{\partial(\rho'H_s)}{\partial t} + \frac{\partial(\rho v_i H_s)}{\partial x_i} = \underbrace{H_s \left(\frac{\partial \rho'}{\partial t} + \frac{\partial(\rho v_i)}{\partial x_i} \right)}_{\text{So, 0 everywhere}} + (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|$$

Outside flow
 $\frac{\partial \rho'}{\partial t} + \frac{\partial(\rho v_i)}{\partial x_i} = 0$ Continuity eq. (in flow region)

Continuity eq. →

$$\frac{\partial(\rho'H_s)}{\partial t} + \frac{\partial(\rho v_i H_s)}{\partial x_i} = (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|$$

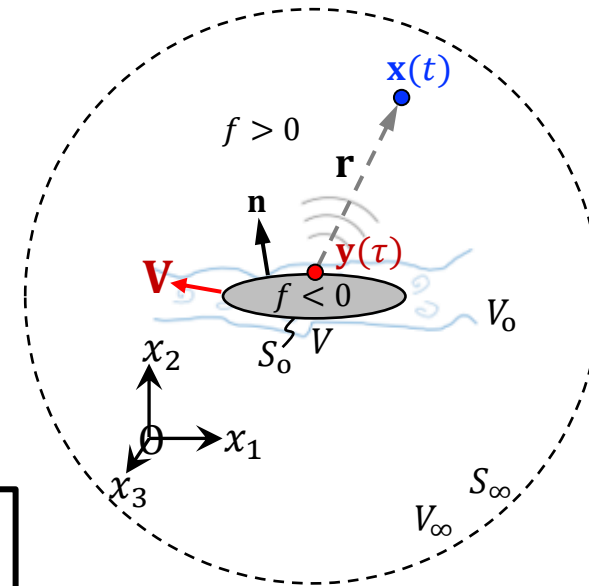
similarly...

Momentum eq. →

$$\frac{\partial(\rho v_i H_s)}{\partial t} + \frac{\partial(\rho v_i v_j H_s + p_{ij} H_s)}{\partial x_j} = (\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f|$$

$$p_{ij} = (p - p_\infty) \delta_{ij} - \sigma_{ij}$$

$$\begin{aligned} \frac{\partial H_s(f)}{\partial x_i} &= \frac{\partial H_s}{\partial f} \nabla f = \delta(f) |\nabla f| \mathbf{n} \\ \frac{\partial H_s(f)}{\partial t} &= \frac{\partial H_s}{\partial f} \frac{\partial f}{\partial t} = \delta(f) \frac{\partial f}{\partial t} \\ \frac{\partial f}{\partial t} &= -\mathbf{V} \cdot \mathbf{n} |\nabla f| \\ \rho' &= \rho - \rho_\infty \end{aligned}$$



The FW-H equation

- To obtain a wave equation, follow the same procedure as for Lighthill's equation...

- Continuity eq.
$$\frac{\partial}{\partial t} \left(\frac{\partial(\rho' H_s)}{\partial t} + \frac{\partial(\rho v_i H_s)}{\partial x_i} \right) = \frac{\partial}{\partial t} \left((\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \right)$$
- Momentum eq.
$$\frac{\partial}{\partial x_i} \left(\frac{\partial(\rho v_i H_s)}{\partial t} + \frac{\partial(\rho v_i v_j H_s + p_{ij} H_s)}{\partial x_j} \right) = \frac{\partial}{\partial x_i} \left((\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \right)$$

(–)
Subtract

Take the result and subtract $\partial^2(\rho' c_\infty^2 H_s) / \partial x_i^2$

➔

$$\frac{\partial^2(\rho' H_s)}{\partial t^2} - c_\infty^2 \frac{\partial^2(\rho' H_s)}{\partial x_i^2} = \frac{\partial^2(T_{ij} H_s)}{\partial x_i \partial x_j} - \frac{\partial}{\partial x_i} \{ (\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \} + \frac{\partial}{\partial t} \{ (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \}$$

FW-H
equation

Lighthill's stress tensor $T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$

Solution to FW-H analogy

- ❑ The method of Green's functions is used to solve the FW-H analogy (as before)
- ❑ We start by writing FW-H analogy in terms of $\mathbf{y}$ and τ as follows:

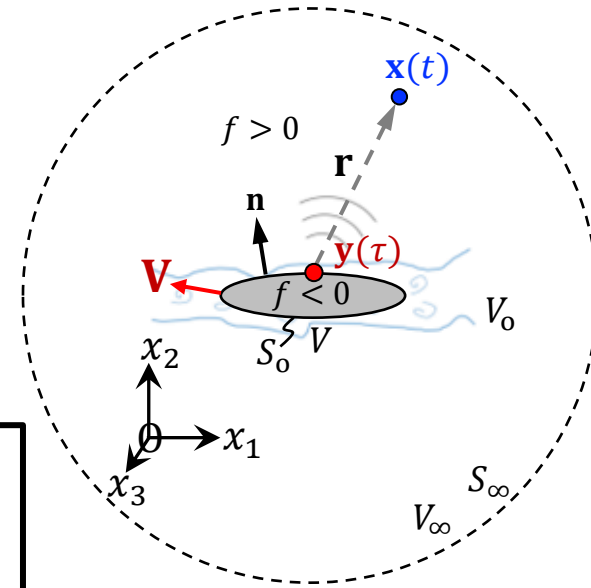
$$\frac{\partial^2(\rho' H_s)}{\partial \tau^2} - c_\infty^2 \frac{\partial^2(\rho' H_s)}{\partial \mathbf{y}_i^2} = \mathcal{S} = \frac{\partial^2(T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} - \frac{\partial}{\partial \mathbf{y}_i} \{(\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f|\} + \frac{\partial}{\partial \tau} \{(\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|\}$$

- ❑ Since the dependent variable ($\rho' H_s$) is **defined in an unbounded medium**, we can ignore the surface integrals (*no difference* between surface BCs and G) and can just use a **simple convolution** ($\int G \mathcal{S}$):

$$\rho'(\mathbf{x}, t) c_\infty^2 H_s = \int_{-T}^T \int_{V_\infty} G \left(\frac{\partial^2(T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau - \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial \mathbf{y}_i} \{(\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f|\} \right) dV(\mathbf{y}) d\tau + \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial \tau} \{(\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|\} \right) dV(\mathbf{y}) d\tau$$

➤ Providing $\mathbf{x}$ is inside V and $-T < t < T$

Lighthill's stress tensor $T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$



Solution to FW-H analogy

$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 H_s = & \int_{-T}^T \int_{V_\infty} G \left(\frac{\partial^2 (T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau \\ & - \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial \mathbf{y}_i} \{ (\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \} \right) dV(\mathbf{y}) d\tau \\ & + \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial \tau} \{ (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \} \right) dV(\mathbf{y}) d\tau \end{aligned}$$

□ We recast the first integral, same as we did in Curle's theorem, using the expansion (Doak's approach):

$$\frac{\partial}{\partial \mathbf{y}_i} \left(G \frac{\partial (T_{ij} H_s)}{\partial \mathbf{y}_j} \right) - \frac{\partial}{\partial \mathbf{y}_i} \left(T_{ij} H_s \frac{\partial G}{\partial \mathbf{y}_i} \right) = G \frac{\partial^2 (T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} - (T_{ij} H_s) \frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j}$$

Substitute into volume integral
& apply divergence theorem

$$\int_{-T}^T \int_{V_\infty} G \left(\frac{\partial^2 (T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau$$

$$\Rightarrow \int_{-T}^T \int_{V_\infty} G \left(\frac{\partial^2 (T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau = \int_{-T}^T \int_{V_\infty} \frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j} (T_{ij} H_s) dV(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left(T_{ij} H_s \frac{\partial G}{\partial \mathbf{y}_i} - G \frac{\partial (T_{ij} H_s)}{\partial \mathbf{y}_i} \right) n_j dS(\mathbf{y}) d\tau$$

Surface integral is 0,
(unbounded domain)

Solution to FW-H analogy

□ Let us expand the second volume integral:

$$\int_{V_\infty} \left(G \frac{\partial}{\partial \mathbf{y}_i} (A_{ij} n_j \delta(f) |\nabla f|) \right) dV(\mathbf{y}) = \underbrace{\int_{V_\infty} \frac{\partial}{\partial \mathbf{y}_i} (G A_{ij} n_j \delta(f) |\nabla f|) dV(\mathbf{y})}_{= - \int_{S_\infty} (G A_{ij} n_j \delta(f) |\nabla f|) \cdot \mathbf{n} dS = 0} - \int_{V_\infty} (A_{ij} n_j \delta(f) |\nabla f|) \frac{\partial G}{\partial \mathbf{y}_i} dV(\mathbf{y})$$

$A_{ij} = (\rho v_i (v_j - V_j) + p_{ij})$

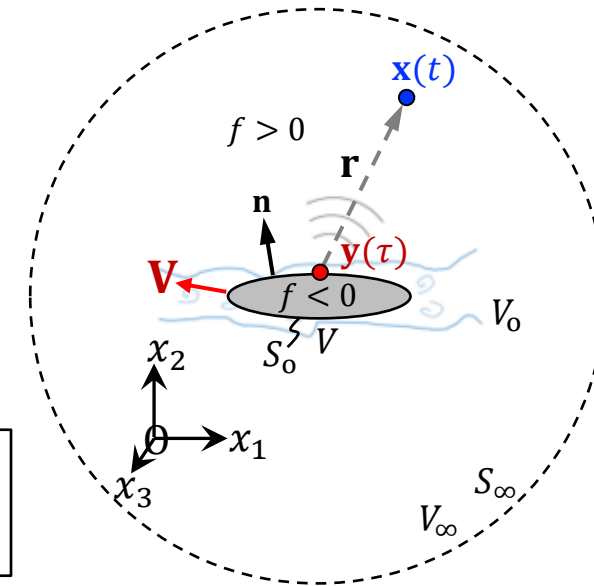
Sommerfeld radiation condition!

$$\Rightarrow \int_{V_\infty} \left(G \frac{\partial}{\partial \mathbf{y}_i} \{A_{ij} n_j \delta(f) |\nabla f|\} \right) dV(\mathbf{y}) = - \int_{V_\infty} (A_{ij} n_j \delta(f) |\nabla f|) \frac{\partial G}{\partial \mathbf{y}_i} dV(\mathbf{y})$$

□ Similarly, for the third volume integral with the time derivative, we can show that integrating by parts and noting that G is zero when $\tau = \pm T$, then:

$$\int_{-T}^T \int_{V_\infty} G \frac{\partial}{\partial \tau} ((\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|) dV(\mathbf{y}) d\tau = - \int_{-T}^T \int_{V_\infty} ((\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|) \frac{\partial G}{\partial \tau} dV(\mathbf{y}) d\tau$$

$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 H_s &= \int_{-T}^T \int_{V_\infty} G \left(\frac{\partial^2 (T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau \\ &\quad - \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial \mathbf{y}_i} \{(\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f|\} \right) dV(\mathbf{y}) d\tau \\ &\quad + \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial \tau} \{(\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|\} \right) dV(\mathbf{y}) d\tau \end{aligned}$$



Solution to FW-H analogy

$$\rho'(\mathbf{x}, t)c_\infty^2 H_s = \int_{-T}^T \int_{V_\infty} \frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j} (T_{ij} H_s) dV(\mathbf{y}) d\tau + \int_{-T}^T \int_{V_\infty} \left((\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \right) \frac{\partial G}{\partial \mathbf{y}_i} dV(\mathbf{y}) d\tau - \int_{-T}^T \int_{V_\infty} \left((\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \right) \frac{\partial G}{\partial \tau} dV(\mathbf{y}) d\tau$$

Diffs. are evaluated at the source

- We introduce the **free-field Green's function**: $G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_\infty - \tau)}{4\pi|\mathbf{x} - \mathbf{y}|}$
 - We can therefore rewrite the solution in terms of G_o :
- Symmetry property $\left\{ \begin{array}{l} \frac{\partial G_o}{\partial \tau} = -\frac{\partial G_o}{\partial t} \\ \frac{\partial G_o}{\partial \mathbf{y}_i} = -\frac{\partial G_o}{\partial \mathbf{x}_i} \\ \frac{\partial^2 G_o}{\partial \mathbf{y}_i \partial \mathbf{y}_j} = \frac{\partial^2 G_o}{\partial \mathbf{x}_i \partial \mathbf{x}_j} \end{array} \right.$

$$\rho'(\mathbf{x}, t)c_\infty^2 H_s = \frac{\partial^2}{\partial \mathbf{x}_i \partial \mathbf{x}_j} \int_{-T}^T \int_{V_\infty} (T_{ij} H_s) G_o dV(\mathbf{y}) d\tau - \frac{\partial}{\partial \mathbf{x}_i} \int_{-T}^T \int_{V_\infty} \left((\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \right) G_o dV(\mathbf{y}) d\tau + \frac{\partial}{\partial t} \int_{-T}^T \int_{V_\infty} \left((\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \right) G_o dV(\mathbf{y}) d\tau$$

Diffs. are evaluated at the observer

Solution to FW-H analogy

- Recall the identity we previously found:

$$\int_{V_\infty} (\mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|) dV = \int_{S_0} \mathbf{v} \cdot \mathbf{n} dS$$

- We use this identity to evaluate:
$$\begin{cases} \int_{-T}^T \int_{V_\infty} ((\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f|) \frac{\partial G}{\partial y_i} dV(\mathbf{y}) d\tau \\ \int_{-T}^T \int_{V_\infty} ((\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|) \frac{\partial G}{\partial \tau} dV(\mathbf{y}) d\tau \end{cases}$$

- Thus, the solution evaluated **at the source** can be expressed as:

$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 = & \int_{-T}^T \int_{V_0(\tau)} \frac{\partial^2 G}{\partial y_i \partial y_j} T_{ij} dV(\mathbf{y}) d\tau + \int_{-T}^T \int_{S_0(\tau)} \frac{\partial G}{\partial y_i} ((\rho v_i (v_j - V_j) + p_{ij}) n_j) dS(\mathbf{y}) d\tau \\ & - \int_{-T}^T \int_{S_0(\tau)} \frac{\partial G}{\partial \tau} ((\rho v_j - \rho' V_j) n_j) dS(\mathbf{y}) d\tau \end{aligned}$$

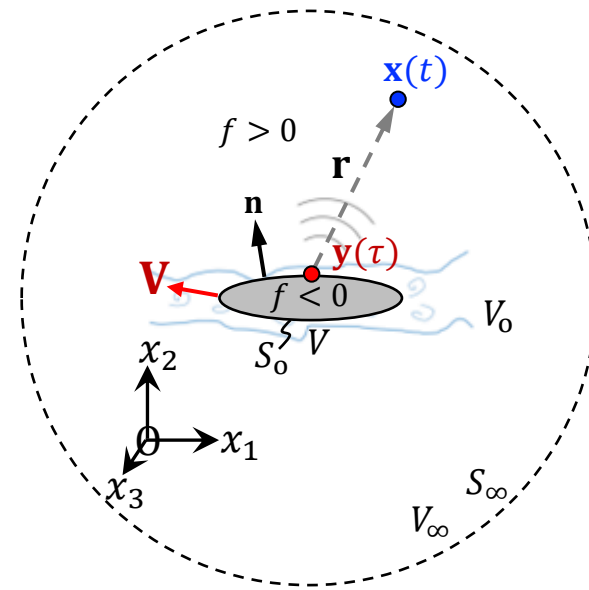
Diffs. are evaluated at the source

- If surface is **stationary** ($\mathbf{V} = 0$), the equation above reduces to **Curle's equation**

- This is NOT an obvious extension due to additional momentum flux term $\rho v_i V_j$, as well as $\rho' V_j$

- **Volume/surface integrals must be evaluated over moving surfaces**

- Both $\mathbf{y}$ and the propagation distance between source and observer ($\mathbf{r}$) **change** with emission time (τ)

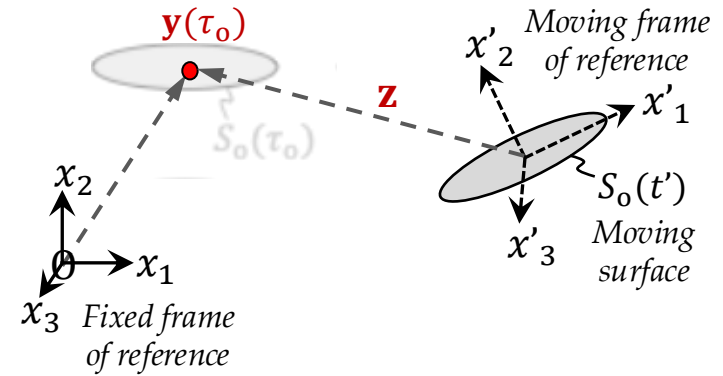


$$\rho'(\mathbf{x}, t) c_\infty^2 = \int_{-T}^T \int_{V_0} \left(\frac{\partial^2 G}{\partial y_i \partial y_j} \right) T_{ij} dV(\mathbf{y}) d\tau + \int_{-T}^T \int_{S_0} \left(\frac{\partial G}{\partial y_i} (\rho v_i v_j + p_{ij}) + G \frac{\partial (\rho v_j)}{\partial \tau} \right) n_j dS(\mathbf{y}) d\tau$$

Solution to FW-H analogy

- ❑ To evaluate the volume/surface integrals, we define a coordinate system $\mathbf{z}$, which moves with the surface
- ❑ Moving coordinate $\mathbf{z}$ is identical to the fixed source coordinate $\mathbf{y}$ at time τ_0 , so for emission times $\tau > \tau_0$ we have:

$$\mathbf{y} = \mathbf{z} + \int_{\tau_0}^{\tau} \mathbf{V}(t') dt'$$



- ❑ Surface/volume integrals can be evaluated in the moving coordinate system $\mathbf{z}$, provided we account for the change in volume/surface elements introduced by the transformation:

$$dV(\mathbf{y}) = \mathbf{J}dV(\mathbf{z}) \quad dS(\mathbf{y}) = \mathbf{K}dS(\mathbf{z}) \quad \mathbf{J} \text{ and } \mathbf{K} \text{ are the Jacobians}$$

- ❑ For surfaces that are moving with *constant linear/angular velocity* (Jacobians are *unity*):

$$dV(\mathbf{y}) = dV(\mathbf{z}) \quad dS(\mathbf{y}) = dS(\mathbf{z}) \quad \frac{\partial G}{\partial \mathbf{y}_i} = \frac{\partial G}{\partial \mathbf{z}_i}$$

Solution to FW-H analogy

- FW-H equation is often applied in **propeller/rotor noise**, where surfaces are rotating and translating in an otherwise **stationary medium**
- As we are concerned with the acoustic far field, the **free-field Green's function** is appropriate:

$$G_o = \frac{\delta(t - \tau - r(\tau)/c_\infty)}{4\pi r(\tau)} \quad r(\tau) = |\mathbf{x} - \mathbf{y}(\tau)| \quad r(\tau) = |\mathbf{x} - \mathbf{z} - [\mathbf{U}_s + (\mathbf{z} \times \boldsymbol{\Omega})](\tau - \tau_o)| \quad \mathbf{y} = \mathbf{z} + \int_{\tau_o}^{\tau} \mathbf{V}(t') dt'$$

↓ **Translational velocity** ↓ **Angular velocity**
(about the origin of $\mathbf{z}$)

- By using the identity: $\int_{V_\infty} (\mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|) dV = \int_{S_o} \mathbf{v} \cdot \mathbf{n} dS$

we can convert volume integrals to surface integrals in the solution evaluated **at the observer**:

$$\rho'(\mathbf{x}, t) c_\infty^2 = \frac{\partial^2}{\partial x_i \partial x_j} \int_{-T}^T \int_{V_\infty} T_{ij} G_o dV(\mathbf{z}) d\tau - \frac{\partial}{\partial x_i} \int_{-T}^T \int_{S_o(\tau)} \left((\rho v_i (v_j - V_j) + p_{ij}) n_j \right) G_o dS(\mathbf{z}) d\tau$$

$$+ \frac{\partial}{\partial t} \int_{-T}^T \int_{S_o(\tau)} \left((\rho v_j - \rho' V_j) n_j \right) G_o dS(\mathbf{z}) d\tau$$

Diffs. are evaluated at the observer

Solution to FW-H analogy

$$\rho'(\mathbf{x}, t) c_\infty^2 = \frac{\partial^2}{\partial x_i \partial x_j} \int_{-T}^T \int_{V_\infty} T_{ij} G_o dV(\mathbf{z}) d\tau - \frac{\partial}{\partial x_i} \int_{-T}^T \int_{S_o(\tau)} ((\rho v_i (v_j - V_j) + p_{ij}) n_j) G_o dS(\mathbf{z}) d\tau + \frac{\partial}{\partial t} \int_{-T}^T \int_{S_o(\tau)} ((\rho v_j - \rho' V_j) n_j) G_o dS(\mathbf{z}) d\tau$$

- Since G_o is multiplied into each integrand, we can now perform the time integrations
- To make it easier, we note that:

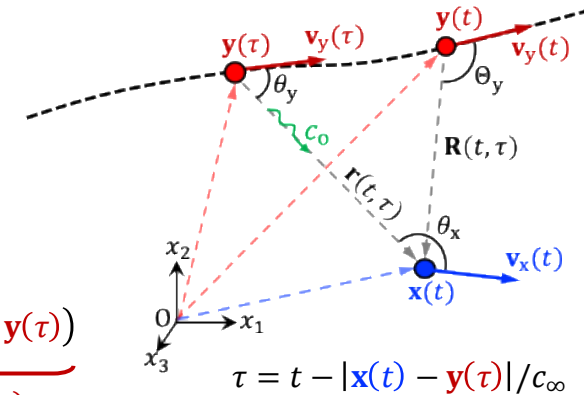
$$g = t - \tau - r(\tau)/c_\infty$$

$$\frac{\partial g}{\partial \tau} = -1 - \frac{1}{c_\infty} \frac{\partial r}{\partial \tau} = -1 + M_r$$

➤ **Source velocity** in the direction of the **observer**:

$$M_r c_\infty = -\frac{\partial r}{\partial \tau}$$

$$\frac{\partial r}{\partial \tau} = \hat{\mathbf{n}}_r \cdot \frac{\partial}{\partial \tau} (\mathbf{x}(t) - \mathbf{y}(\tau)) = -\mathbf{v}_y(\tau)$$



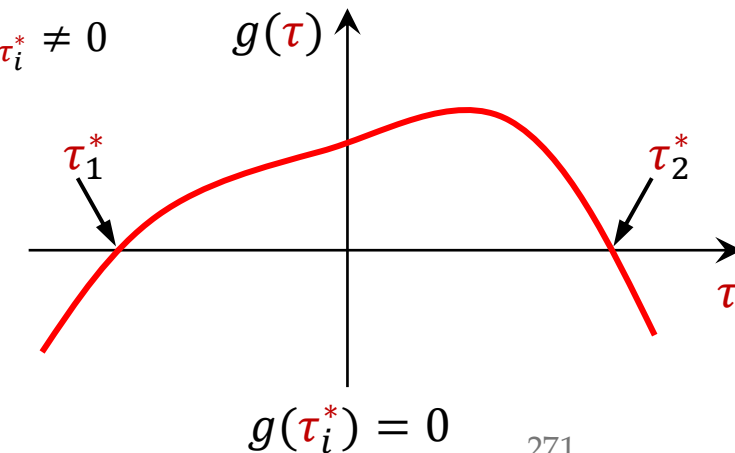
- In addition, recall the **composition** feature of the Dirac delta function:

- $g(\tau)$ – smooth function (continuously differentiable)
- $g(\tau)$ has n number of zeros $\tau_i^* = [\tau_1^*, \dots, \tau_n^*]$, such that $g(\tau_i^*) = 0$ and $[\partial g / \partial \tau]_{\tau_i^*} \neq 0$

$$\int_{-T}^T f(\tau) \delta(g(\tau)) d\tau = \int_{g(-T)}^{g(T)} \frac{f(\tau) \delta(g)}{\partial g / \partial \tau} dg \Rightarrow = \sum_{i=1}^n \left[\frac{f}{|\partial g / \partial \tau|} \right]_{\tau_i^*} = \sum_{i=1}^n \left[\frac{f}{|1 - M_r|} \right]_{\tau_i^*}$$

- Correct retarded time τ^*** is the solution to:

$$\tau^* = t - r(\tau^*)/c_\infty$$



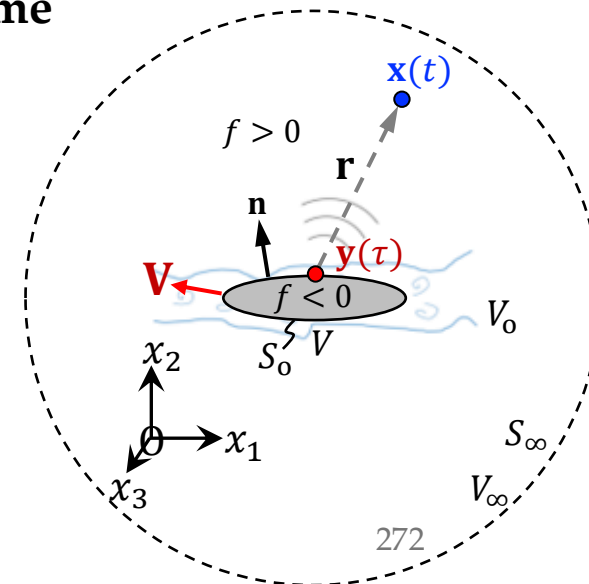
FW-H equation

General solution using G_o

□ Therefore, the Ffowcs Williams and Hawkins equation becomes:

$$\rho'(\mathbf{x}, t) c_\infty^2 H_s = \frac{\partial^2}{\partial x_i \partial x_j} \int_{V_o} \left[\frac{T_{ij}}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dV(\mathbf{z}) - \frac{\partial}{\partial x_i} \int_{S_o} \left[\frac{(\rho v_i (v_j - V_j) + p_{ij}) n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z}) + \frac{\partial}{\partial t} \int_{S_o} \left[\frac{(\rho v_j - \rho' V_j) n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z})$$

□ Integrals MUST be carried over the **moving surface** at the **correct emission time**
 $\tau^* = t - r(\tau^*)/c_\infty$

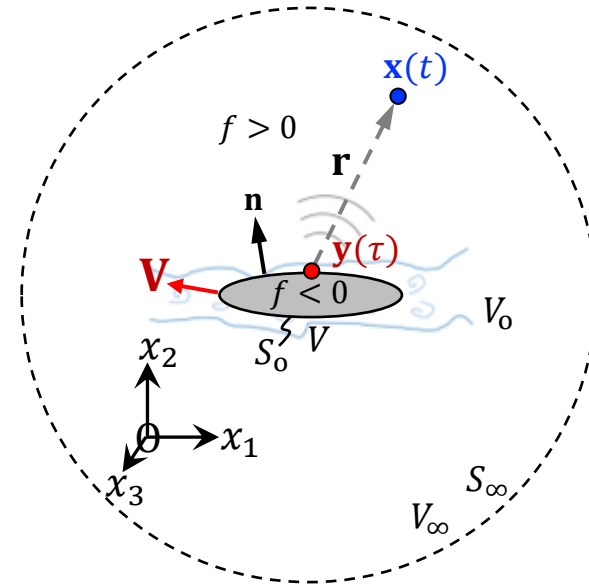


FW-H equation

Far-field approximation

- In the **acoustic far field**, space and time derivatives can be exchanged
 - Sound is the only disturbance

A change in acoustic pressure **in time** at a fixed point is the same as the change **in space** multiplied by the sound speed



$$\frac{\partial G_o}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\delta(g)}{4\pi r(\tau)} \right) = \frac{\partial r}{\partial x_i} \frac{\partial}{\partial r} \left(\frac{\delta(g)}{4\pi r(\tau)} \right) = \frac{\partial r}{\partial x_i} \left(\frac{\partial g}{\partial r} \frac{\partial \delta(g)}{\partial g} \frac{1}{4\pi r(\tau)} - \frac{\delta(g)}{4\pi r^2(\tau)} \right) = \frac{x_i - y_i}{r} \left(-\frac{1}{c_\infty} \frac{\partial \delta(g)}{\partial g} \frac{1}{4\pi r(\tau)} - \frac{\delta(g)}{4\pi r^2(\tau)} \right)$$

$$\frac{\partial G_o}{\partial x_i}$$

$$= -\frac{x_i - y_i}{r} \left(\frac{1}{c_\infty} \frac{\partial G_o}{\partial t} + \frac{G_o}{r(\tau)} \right)$$

And for the **acoustic far-field** $|\mathbf{x}| \gg |\mathbf{y}|$:

$$\frac{\partial G_o}{\partial x_i} \approx -\frac{x_i}{|\mathbf{x}|} \left(\frac{1}{c_\infty} \frac{\partial G_o}{\partial t} \right)$$

$$G_o = \frac{\delta(g)}{4\pi r(\tau)}$$

$$g = t - \tau - r(\tau)/c_\infty$$

$$r = |\mathbf{x} - \mathbf{y}|$$

$$\frac{\partial r}{\partial x_i} = \frac{x_i - y_i}{r}$$

FW-H equation

Far-field approximation

- Introducing this change into the **dipole** and **quadrupole** terms gives the **far-field approximation of the FW-H equation**:

$$\rho'(\mathbf{x}, t)c_\infty^2 \approx \frac{x_i x_j}{|\mathbf{x}|^2} \frac{1}{c_\infty^2} \frac{\partial^2}{\partial t^2} \int_{V_0} \left[\frac{T_{ij}}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dV(\mathbf{z}) + \frac{x_i}{|\mathbf{x}|} \frac{1}{c_\infty} \frac{\partial}{\partial t} \int_{S_0} \left[\frac{(\rho v_i (v_j - V_j) + p_{ij}) n_j}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z})$$

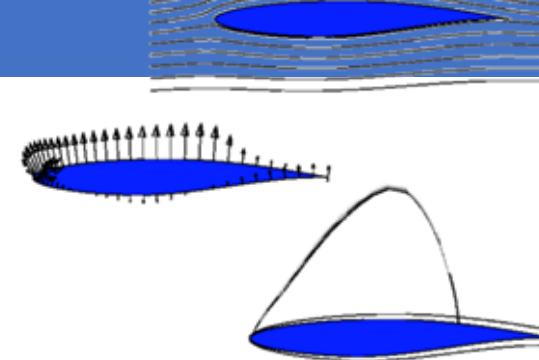
$$+ \frac{\partial}{\partial t} \int_{S_0} \left[\frac{(\rho v_j - \rho' V_j) n_j}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z})$$

- All the integrands are evaluated at **emission time** $\tau^* = t - r(\tau^*)/c_\infty \Rightarrow$ **Emitted time history** and the **observed time history** are *distorted* relative to each other by the motion
- In applications where the source terms are defined as a function of source time (e.g., rotors), it is useful to shift from observer time to source time using:

$$\frac{\partial}{\partial t} = \frac{\partial \tau^*}{\partial t} \frac{\partial}{\partial \tau^*} = \frac{1}{1 - M_r} \frac{\partial}{\partial \tau^*}$$

FW-H equation

Far-field approximation for impenetrable surfaces



□ **Far-field approximation of the FW-H equation for *impenetrable* surfaces:**

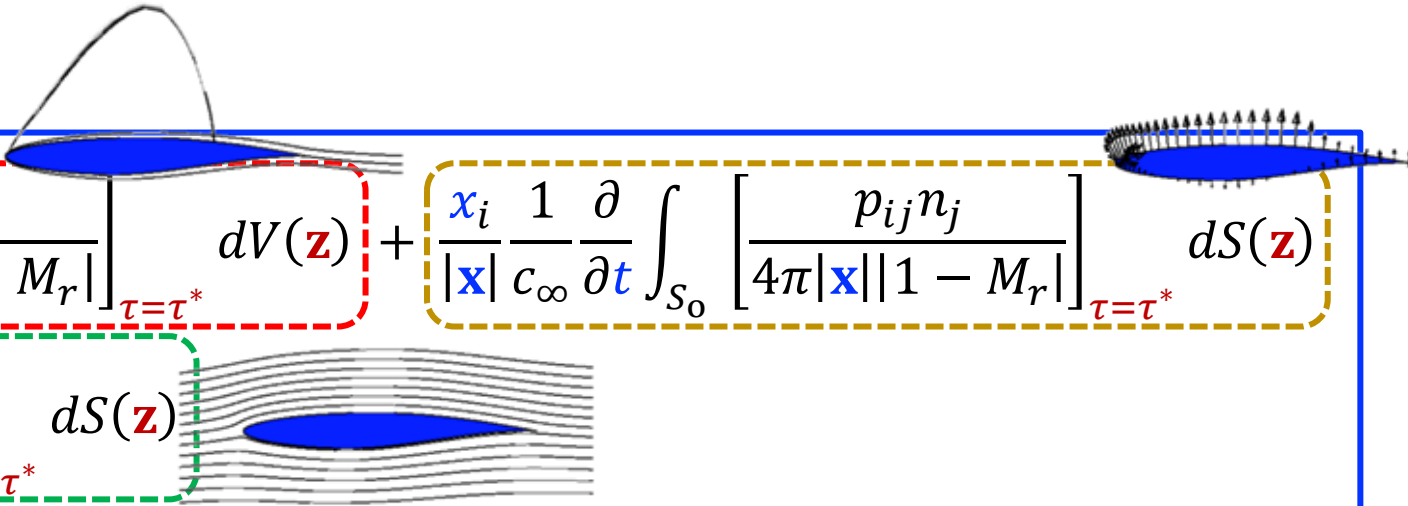
➤ Flow velocity normal to the surface equals the **surface** normal velocity - $v_i n_i = V_i n_i$

$$\rho - \rho' = \rho_\infty$$

$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 \approx & \frac{x_i x_j}{|\mathbf{x}|^2} \frac{1}{c_\infty^2} \frac{\partial^2}{\partial t^2} \int_{V_0} \left[\frac{T_{ij}}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dV(\mathbf{z}) + \frac{x_i}{|\mathbf{x}|} \frac{1}{c_\infty} \frac{\partial}{\partial t} \int_{S_0} \left[\frac{p_{ij} n_j}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z}) \\ & + \frac{\partial}{\partial t} \int_{S_0} \left[\frac{\rho_\infty V_j n_j}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z}) \end{aligned}$$

FW-H equation

Application for rotors/propellers

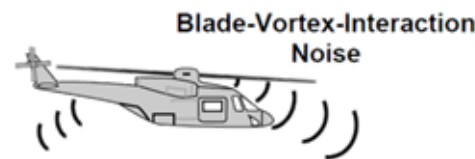
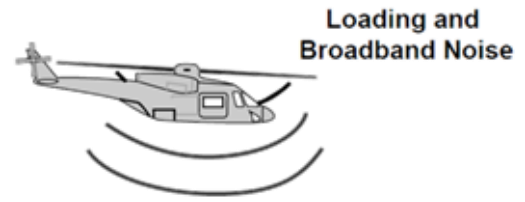
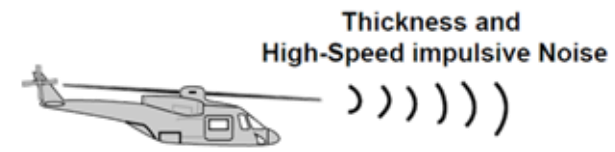
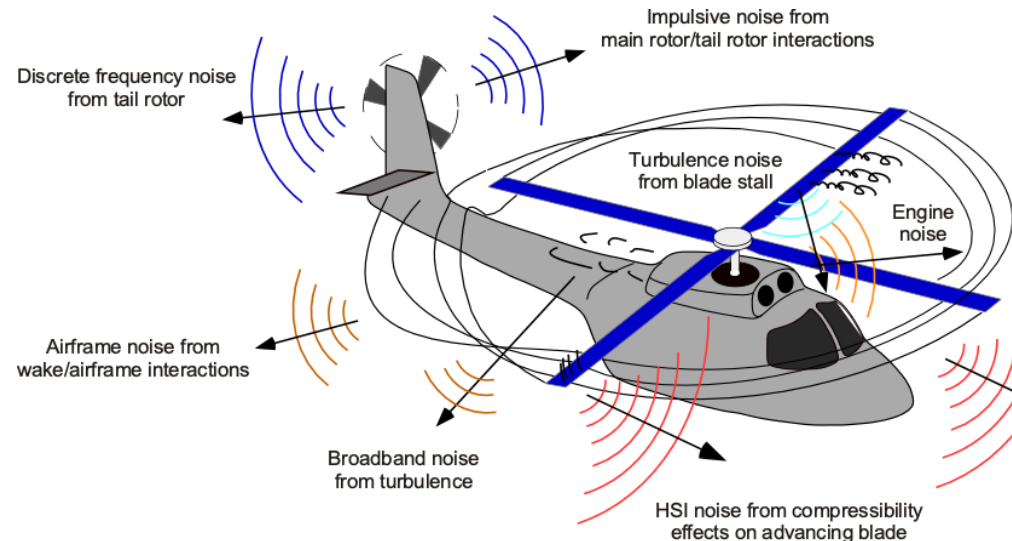
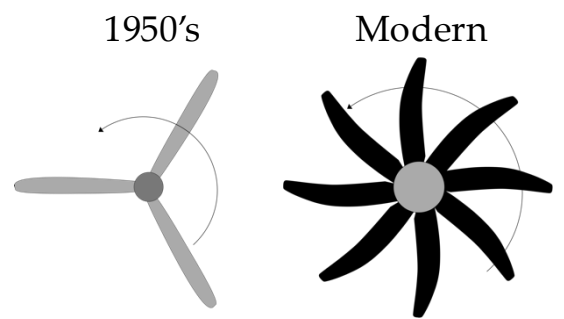


$$\rho'(\mathbf{x}, t) c_\infty^2 \approx \underbrace{\frac{x_i x_j}{|\mathbf{x}|^2} \frac{1}{c_\infty^2} \frac{\partial^2}{\partial t^2} \int_{V_0} \left[\frac{T_{ij}}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dV(\mathbf{z})}_{\text{Quadrupole noise}} + \underbrace{\frac{x_i}{|\mathbf{x}|} \frac{1}{c_\infty} \frac{\partial}{\partial t} \int_{S_0} \left[\frac{p_{ij} n_j}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z})}_{\text{Loading (dipole) noise}} + \underbrace{\frac{\partial}{\partial t} \int_{S_0} \left[\frac{\rho_\infty V_j n_j}{4\pi |\mathbf{x}| |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z})}_{\text{Thickness (monopole) noise}}$$

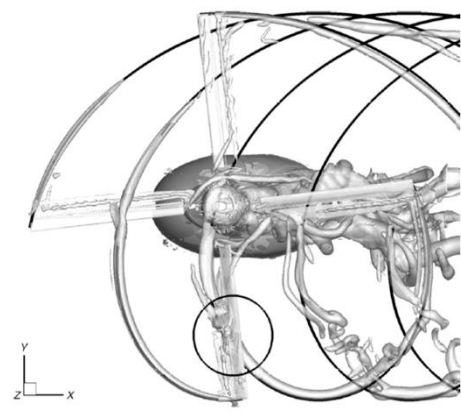
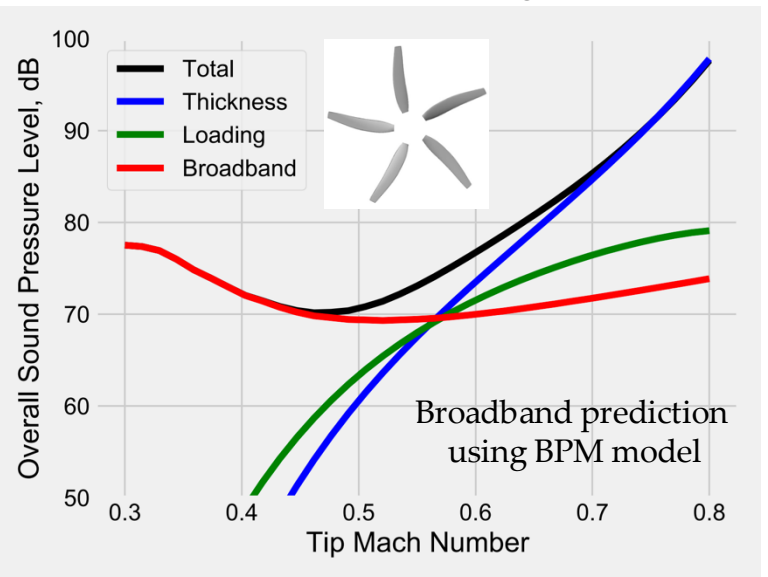
- Three main noise sources are recognized for *rotating blades with impenetrable blades*:
- **Quadrupole noise** (T_{ij}) – due to **turbulence** and **flow distortions** (e.g., shock waves)
 - **Loading (dipole) noise** ($p_{ij} n_j$) – due to **surface loading (force)**
 - **Thickness (monopole) noise** ($\rho_\infty V_j n_j$) – due to a **volume displacement** source arising when the observer “sees” a time varying surface velocity at emission time
 - Zero for an object in linear motion traveling towards the observer, but non-zero for a rotating blade that continuously changes direction relative to the observer

FW-H equation

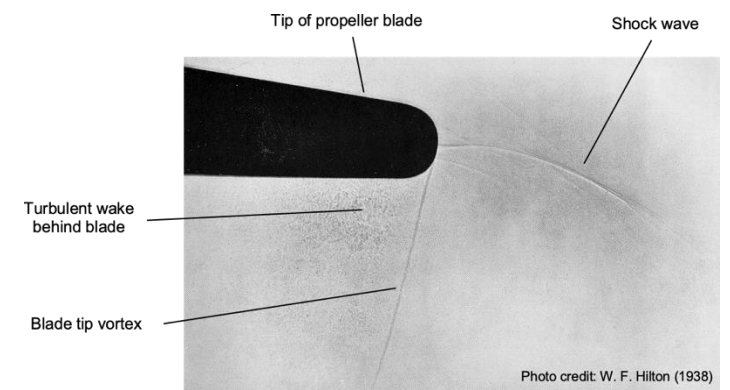
Application for rotors/propellers



eVTOL rotor FW-H prediction
(thickness + loading)



Aircraft Aeroacoustics



Moving sources

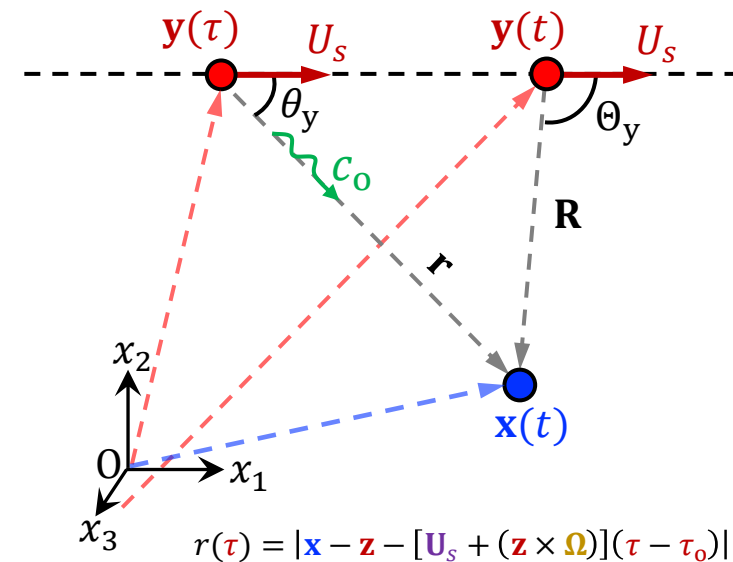
- Consider an acoustically compact source moving at constant speed U_s
 - Ignore sound from turbulence ($T_{ij} \approx 0$) and assume stationary medium
- Distance to observer is given by: $r(\tau) = |\mathbf{x} - \mathbf{y}^{(c)}| = |\mathbf{x} - U_s \tau|$
 - $\mathbf{y}^{(c)}$ - centroid of the surface, at the correct retarded time, in the fixed reference frame, corresponding to $\mathbf{z} = 0$ in the moving frame
 - We take the time origin of the surface motion $\tau_0 = 0$
- Applying FW-H equation for an impenetrable surface:

$$\rho'(\mathbf{x}, t) c_\infty^2 \approx - \frac{\partial}{\partial x_i} \int_{S_0} \left[\frac{p_{ij} n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z}) + \frac{\partial}{\partial t} \int_{S_0} \left[\frac{\rho_\infty V_j n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS(\mathbf{z})$$

- But since the source is acoustically compact, we can take the propagation factor outside of the integrals:

$$\rho'(\mathbf{x}, t) c_\infty^2 \approx - \frac{\partial}{\partial x_i} \left[\frac{1}{4\pi r |1 - M_r|} \int_{S_0} p_{ij} n_j dS(\mathbf{z}) \right]_{\tau=\tau^*} + \frac{\partial}{\partial t} \left[\frac{1}{4\pi r |1 - M_r|} \int_{S_0} \rho_\infty V_j n_j dS(\mathbf{z}) \right]_{\tau=\tau^*}$$

=0 for any
closed rigid body



Moving sources

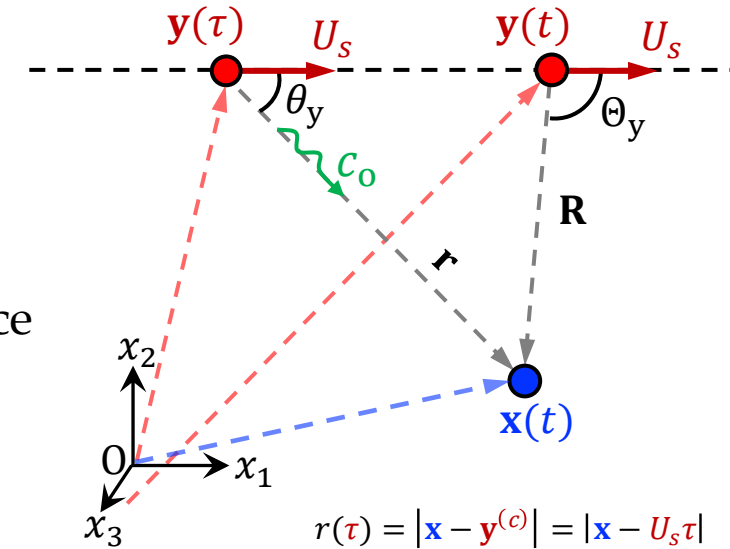
- ❑ Thus, only **surface loading** will radiate sound
- ❑ For modelling, assume a harmonic dipole source in the frame of reference moving with the surface

$\hat{F}_i$ - vector force amplitude

$$\int_{S_0} p_{ij} n_j dS(\mathbf{z}) = \mathcal{R}e(\hat{F}_i e^{-i\omega_0 \tau})$$

- ❑ The **acoustic field** is therefore:

$$\rho'(\mathbf{x}, t) c_\infty^2 \approx -\mathcal{R}e \left(\frac{\partial}{\partial x_i} \left[\frac{\hat{F}_i e^{-i\omega_0 \tau}}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} \right)$$



Moving sources

- In the **far-field**, retarded time equation can be approximated with $y_1 = U_s \tau$, yielding:

$$r(\tau) \approx |\mathbf{x}| - \frac{x_1 U_s \tau}{|\mathbf{x}|} + \dots$$

- Approximation only applies over short periods ($U_s \ll |\mathbf{x}|$), such that $x_1/|\mathbf{x}| = \mathbb{C}$

- **Velocity** and **Mach number** of the **source** in the direction of the observer:

$$U_r = \frac{U_s x_1}{|\mathbf{x}|} \quad M_r = \frac{U_r}{c_\infty}$$

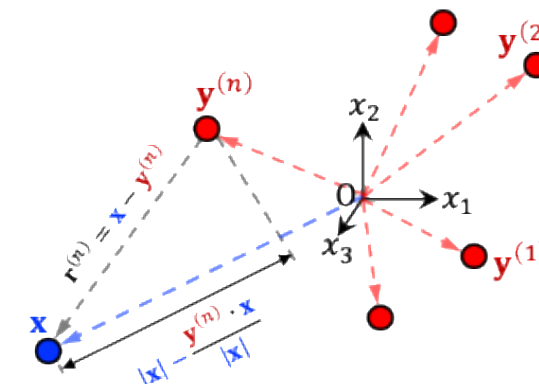
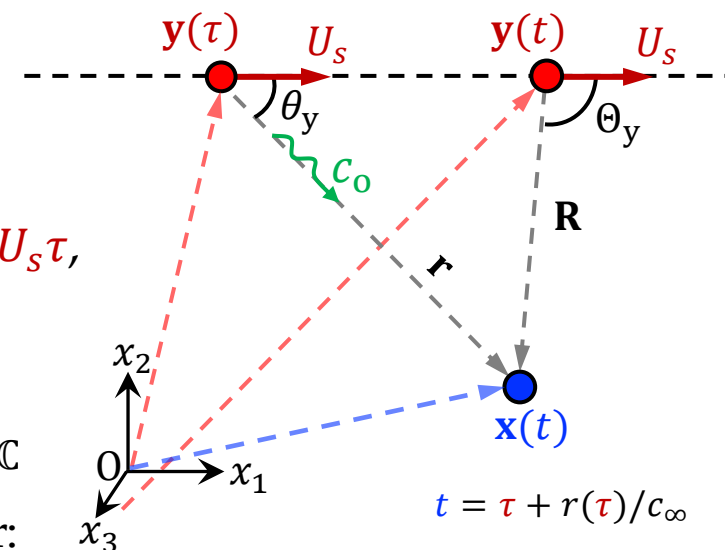
- The far field approximation applied then gives: $t - \tau \approx |\mathbf{x}|/c_\infty - M_r \tau$ or $\tau \approx (t - |\mathbf{x}|/c_\infty)/(1 - M_r)$

- Therefore, the **far-field** approximation gives:

Ignoring near field terms of order $|\mathbf{x}|^{-2} \Rightarrow \rho'(\mathbf{x}, t) c_\infty^2 \approx -\text{Re} \left(\frac{\partial}{\partial x_i} \left[\frac{\hat{F}_i e^{-i\omega_0(t - |\mathbf{x}|/c_\infty)/(1 - M_r)}}{4\pi |\mathbf{x}| (1 - M_r)} \right] \right)$

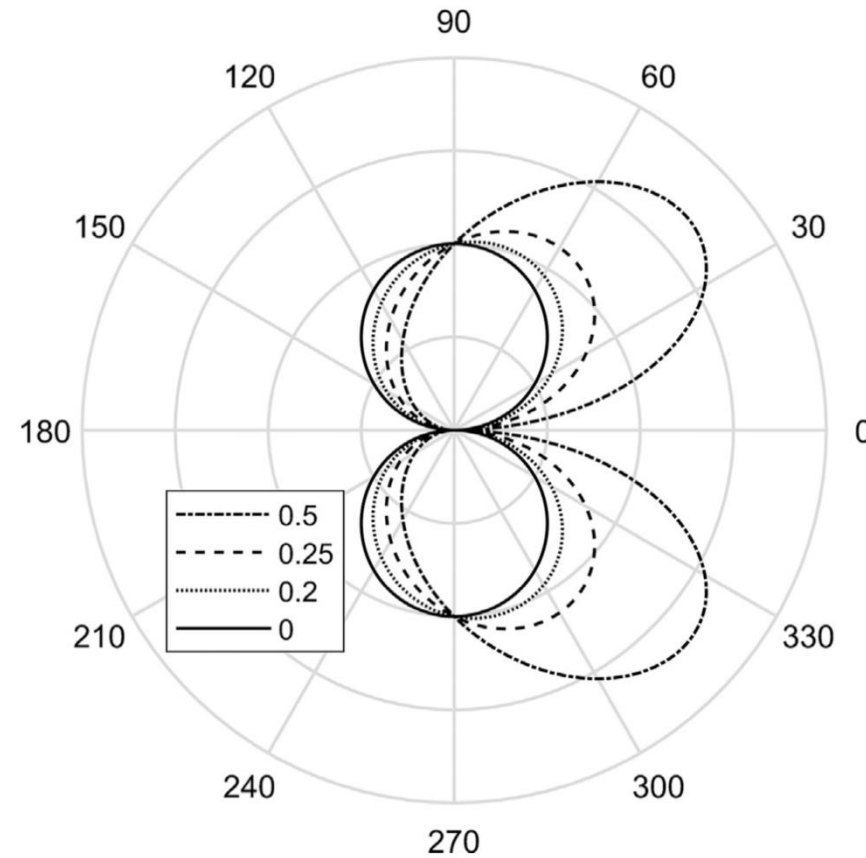
$$\rho'(\mathbf{x}, t) c_\infty^2 \approx \text{Re} \left(\frac{-i\omega_0 x_i \hat{F}_i e^{-i\omega_0(t - |\mathbf{x}|/c_\infty)/(1 - M_r)}}{4\pi c_\infty |\mathbf{x}|^2 (1 - M_r)^2} + \mathcal{O}(|\mathbf{x}|^{-2}) \right)$$

Doppler frequency shift



$$r = |\mathbf{x} - \mathbf{y}| \approx |\mathbf{x}| - \frac{\mathbf{y} \cdot \mathbf{x}}{|\mathbf{x}|}$$

Moving sources



Directivity of a dipole source with an axis perpendicular to its direction of motion as a function of source Mach number

Moving sources

- Retarded time is evaluated by solving $t = \tau + r(\tau)/c_\infty$, or:

$$t - \tau = \left((x_1 - U_s \tau)^2 + x_2^2 + x_3^2 \right)^{1/2} / c_\infty$$

- Retarded time $\tau = \tau^*$ can be obtained in terms of the observer time by squaring the above equation:

$$c_\infty^2 (t - \tau)^2 = (x_1 - U_s \tau)^2 + x_2^2 + x_3^2$$



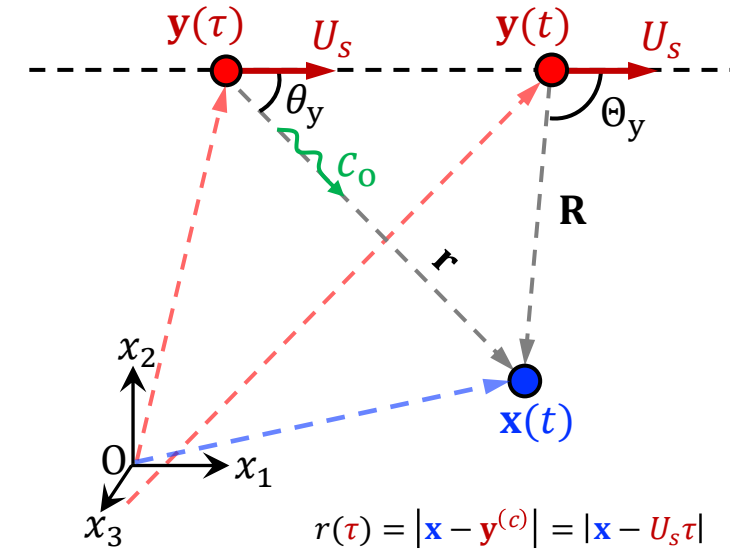
$$\tau^* = \frac{c_\infty t - M x_1 \pm \sqrt{(x_1 - U_s t)^2 + (1 - M^2)(x_2^2 + x_3^2)}}{c_\infty (1 - M^2)}$$

- Two possible solutions: $\tau_1 (-)$ and $\tau_2 (+)$ | $\tau_2 > \tau_1$

➤ Solution is constrained by **causality condition**: $t > \tau$

- For subsonic flow ($M < 1$) → only τ_1 applies (-)

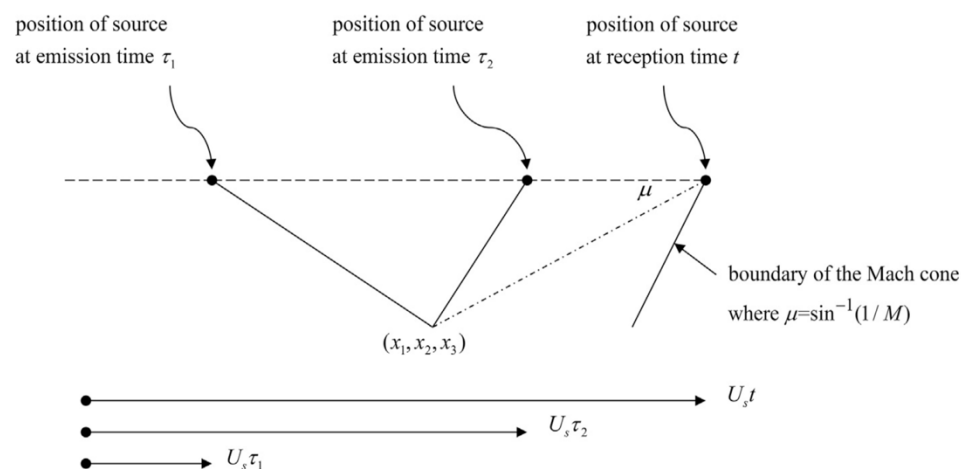
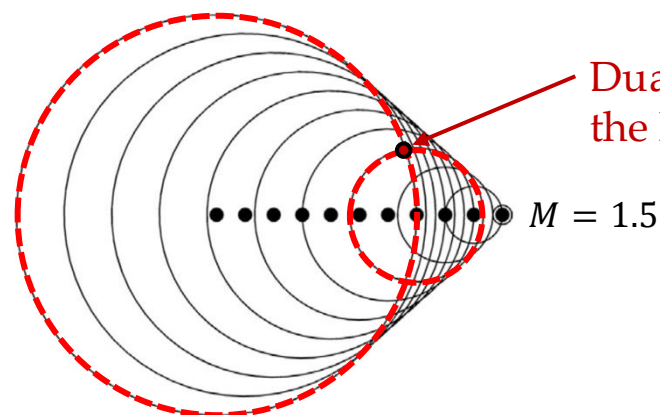
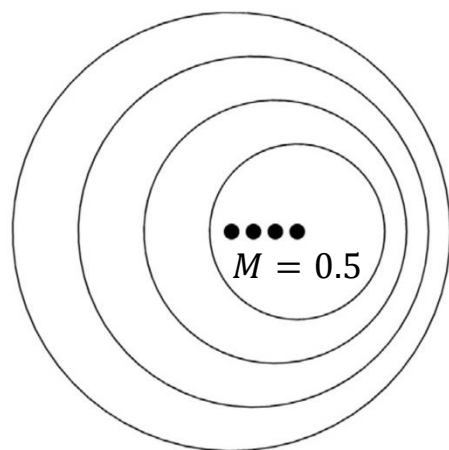
- For supersonic flow ($M > 1$) → two *real* solutions are possible if: $(x_1 - U_s t)^2 > (M^2 - 1)(x_2^2 + x_3^2)$



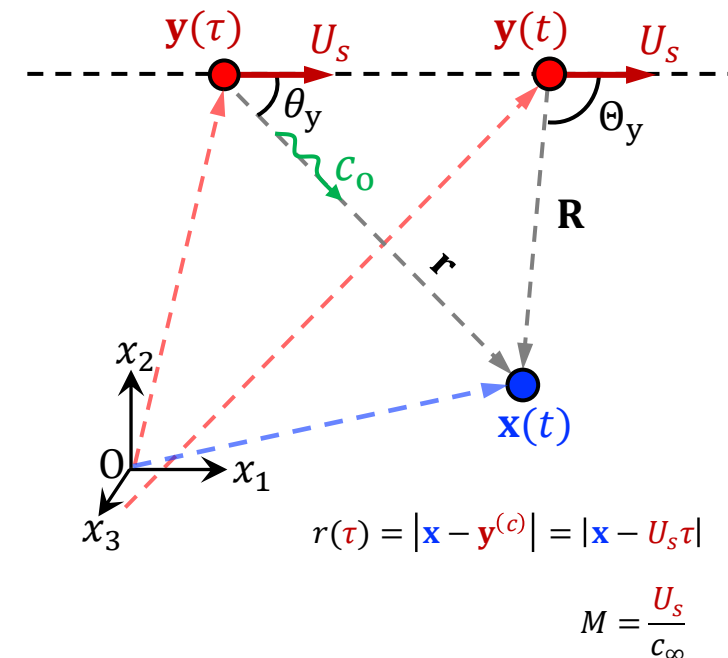
$$M = \frac{U_s}{c_\infty}$$

Moving sources

Wave fronts from a source moving from left to right



Aircraft Aeroacoustics



$$(x_1 - U_s t)^2 > (M^2 - 1)(x_2^2 + x_3^2)$$

$$\Rightarrow \frac{|x_1 - U_s t|}{\sqrt{x_2^2 + x_3^2}} = \cot \mu > \sqrt{M^2 - 1}$$

$$\mu < \sin^{-1} \left(\frac{1}{M} \right)$$

Mach cone angle

μ – angle of the observer to the path of the source evaluated at reception time